Transcendence in the Affine Case of Erdős Problem 270

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Erdős Problem 270 asked whether a series of reciprocals of consecutive-integer products must be irrational when the number of factors tends to infinity. Crmarić and Kovač showed that the answer is negative without a regularity assumption, while the case in which the number of factors is nondecreasing remains open. We prove that

$$C_{a,b} := \sum_{n=1}^{\infty} \frac{n!}{((a+1)n+b)!}$$

is transcendental for every $a \geq 1$ and $b \geq 1-a$. We first give an elementary proof of irrationality for $0 \leq b \leq a$, including the example singled out by Erdős and Graham. We then prove transcendence by studying the associated hypergeometric E-functions. Theorems of Salikhov and Salikhov–Viskina cover $0 \leq b \leq a$, and an index shift handles $1-a \leq b < 0$. For $b > a$, we obtain the functional independence required by Beukers's lifting theorem from an Euler-operator reduction and a logarithmic formal term at infinity.

1 Irrationality of the Erdős–Graham example

Erdős and Graham ended their chapter on irrationality and transcendence with the question now known as Erdős Problem 270. Given a positive integer-valued function f with $f(n) \to \infty$, they considered

$$S(f) := \sum_{n=1}^{\infty} \frac{1}{(n+1)(n+2)\cdots(n+f(n))},$$

and asked whether $S(f)$ must be irrational. They expected an affirmative answer when f is nondecreasing [1, p. 66]. Crmarić and Kovač later showed that some regularity assumption is necessary: as f ranges over all such functions, $S(f)$ takes every value in $(0, \infty)$ [3, Theorem 1]. The nondecreasing problem remains open [2].

Erdős and Graham singled out the simplest nondecreasing choice, $f(n) = n$. They described the resulting series as difficult to treat and wrote that such series were surely irrational [1, pp. 65–66]. Crmarić and Kovač later returned to this example and reported that they knew no proof of its irrationality [3, p. 55, following (1.1)]. After their paper appeared, Crmarić and Kovač found a short proof, which Kovač posted on the Erdős Problems forum in July 2026 [4]. We obtained the argument below independently, before learning of their post. The series is the first member of the affine family considered here:

$$S(f) = C := \sum_{n=1}^{\infty} \frac{n!}{(2n)!} = \sum_{n=1}^{\infty} \frac{1}{D_n}, \qquad D_n := \frac{(2n)!}{n!}.$$

Then

$$\frac{D_{n+1}}{D_n} = \frac{(2n+2)(2n+1)}{n+1} = 4n+2,$$

so $D_n \mid D_{n+1}$. If

$$P_N := \sum_{n=1}^{N} \frac{1}{D_n}, \qquad R_N := C - P_N,$$

then $D_N P_N \in \mathbb{Z}$, while

$$\begin{aligned} 0 < D_N R_N &= \sum_{k=1}^{\infty} \frac{D_N}{D_{N+k}} \\ &= \sum_{k=1}^{\infty} \frac{1}{(4N+2)(4N+6)\cdots(4N+4k-2)} \\ &\le \sum_{k=1}^{\infty} \frac{1}{(4N+2)^k} = \frac{1}{4N+1}. \end{aligned}$$

Thus $D_N R_N \to 0$.

Suppose that $C = p/q$, where $p \in \mathbb{Z}$ and $q \in \mathbb{Z}_{>0}$. Then

$$D_N R_N = \frac{pD_N}{q} - D_N P_N$$

is positive and belongs to $q^{-1}\mathbb{Z}$. It must therefore be at least $1/q$, contradicting $D_N R_N \to 0$. Hence C is irrational.

2 Irrationality for $0 \le b \le a$

For integers $a \ge 1$ and $b \ge 0$, the affine choice $f(n) = an + b$ gives

$$S(f) = \sum_{n=1}^{\infty} \frac{1}{(n+1)\cdots((a+1)n+b)} = \sum_{n=1}^{\infty} \frac{n!}{((a+1)n+b)!} =: C_{a,b}.$$

These are the affine cases of Problem 270 with nonnegative intercept. The preceding argument extends to the range $0 \le b \le a$. Fix a and b in this range, and put

$$D_n := \frac{((a+1)n+b)!}{n!}.$$

Then

$$\frac{D_{n+1}}{D_n} = \frac{\prod_{j=1}^{a+1} \big((a+1)n+b+j\big)}{n+1}.$$

The factor with $j = a+1-b$ is $(a+1)(n+1)$. Cancelling $n+1$ gives the positive integer

$$r_n := \frac{D_{n+1}}{D_n} = (a+1) \prod_{\substack{1 \le j \le a+1 \\ j \ne a+1-b}} \big((a+1)n+b+j\big).$$

Thus $D_n \mid D_{n+1}$. Moreover, $r_n \geq a+1 \geq 2$, and every factor in the product defining r_n increases with n. Hence r_n is increasing and tends to infinity. Let

$$P_N := \sum_{n=1}^{N} \frac{1}{D_n}, \qquad R_N := C_{a,b} - P_N.$$

Since $D_1 \mid D_2 \mid \cdots \mid D_N$, we have $D_N P_N \in \mathbb{Z}$. Moreover,

$$\begin{aligned} 0 < D_N R_N &= \sum_{k=1}^{\infty} \frac{1}{r_N r_{N+1} \cdots r_{N+k-1}} \\ &\leq \sum_{k=1}^{\infty} \frac{1}{r_N^k} = \frac{1}{r_N - 1} \longrightarrow 0. \end{aligned}$$

If $C_{a,b} = p/q$, then $D_N R_N$ is positive and belongs to $q^{-1}\mathbb{Z}$, so it is at least $1/q$. This is impossible for large N. Hence $C_{a,b}$ is irrational for every $a \geq 1$ and $0 \leq b \leq a$.

3 Transcendence of $C_{1,0}$ and $C_{1,1}$

The elementary proof above establishes the irrationality of $C_{1,0}$. Its Gaussian integral representation leads to an independent proof of transcendence. Crmarić and Kovač record the identity [3, p. 55 and footnote 1]

$$C_{1,0} = e^{1/4} \int_0^{1/2} e^{-t^2}\, dt.$$

They obtain it by expanding $e^{1/4-t^2}$ and integrating termwise.

In a 2023 MathOverflow discussion of the neighboring integral $\int_0^1 e^{-t^2}\, dt$, Weber and Pinelis [5] considered the E-function

$$y(x) := \frac{1}{\sqrt{x}} \int_0^{\sqrt{x}} e^{-t^2}\, dt = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} \frac{x^n}{n!}$$

and used the Siegel–Shidlovsky theorem to prove that $y(1)$ and e^{-1} are algebraically independent. The differential system in their argument is ordinary at every nonzero algebraic point, so applying the same theorem at $x = 1/4$ gives the algebraic independence of $y(1/4)$ and $e^{-1/4}$. Since

$$y(1/4) = 2\int_0^{1/2} e^{-t^2}\, dt, \qquad C_{1,0} = \frac{y(1/4)}{2e^{-1/4}},$$

an algebraic value of $C_{1,0}$ would give a nontrivial algebraic relation between $y(1/4)$ and $e^{-1/4}$. Hence $C_{1,0}$ is transcendental. The MathOverflow discussion did not mention either the factorial series or this specialization.

To pass from this special argument to the full family, we express the case $a = 1$ as a hypergeometric E-function. This introduces the method used in Section 4 and treats $C_{1,1}$ as well. For $b \in \{0, 1\}$, define

$$F_{1,b}(z) := \sum_{n=0}^{\infty} \frac{n!}{(2n+b)!} z^n.$$

Then

$$F_{1,b}(1) = 1 + C_{1,b}.$$

Writing $\beta = b + 1/2$, direct coefficient comparison gives

$$F_{1,b}(z) = {}_1F_1\left(1; \beta; \frac{z}{4}\right).$$

This is an E-function. If $h(x) = {}_1F_1(1; \beta; x)$, Kummer's equation reads

$$xh'' + (\beta - x)h' - h = 0.$$

The left side is the derivative of $xh' + (\beta - 1 - x)h$. Evaluation at $x = 0$ therefore gives

$$xh'(x) + (\beta - 1 - x)h(x) = \beta - 1.$$

After substituting $x = z/4$,

$$zF'_{1,b}(z) + \left(\beta - 1 - \frac{z}{4}\right) F_{1,b}(z) = \beta - 1. \tag{1}$$

Thus $(1, F_{1,b})^{\mathsf{T}}$ satisfies a first-order system over $\overline{\mathbb{Q}}(z)$ whose only finite singularity is 0. The functions 1 and $F_{1,b}$ are linearly independent over $\overline{\mathbb{Q}}(z)$. Otherwise $F_{1,b}$ would be rational; because it is entire, it would be a polynomial, contrary to its infinitely many nonzero Taylor coefficients. Beukers [6, Corollary 1.4] proves that E-functions which are linearly independent over $\overline{\mathbb{Q}}(z)$ remain linearly independent after evaluation at a nonzero algebraic point that is ordinary for their differential system. Applied at $z = 1$, this gives linear independence of 1 and $F_{1,b}(1)$ over $\overline{\mathbb{Q}}$. Hence $F_{1,b}(1)$ is transcendental, and so are both $C_{1,0}$ and $C_{1,1}$. The case $b = 0$ recovers the preceding conclusion, while $b = 1$ gives the other case in the range $0 \le b \le a$ when $a = 1$.

4 Transcendence of the affine family

Theorem 1. *For all integers $a \ge 1$ and $b \ge 1 - a$, the number $C_{a,b}$ is transcendental.*

For all integers $a \ge 1$ and $b \ge 0$, define

$$F_{a,b}(z) := b! \sum_{n=0}^{\infty} \frac{n!}{((a+1)n + b)!} z^{an}. \tag{2}$$

Then

$$F_{a,b}(1) = 1 + b! C_{a,b}. \tag{3}$$

For $b \ge 0$, it is therefore enough to prove that $F_{a,b}(1)$ is transcendental. Beukers's theorem reduces this to finding a differential system, ordinary at 1, whose components include $F_{a,b}$ and are linearly independent over $\overline{\mathbb{Q}}(z)$.

4.1 The range $0 \le r \le a$

Assume first that $0 \le r \le a$. Put

$$J_{a,r} := \{1, \dots, a+1\} \setminus \{a+1-r\}, \qquad \beta_j := \frac{r+j}{a+1} \quad (j \in J_{a,r}).$$

The omitted parameter is the unique one equal to 1. The multiplication formula gives

$$F_{a,r}(z) = {}_1F_a\left(\begin{matrix} 1 \\ (\beta_j)_{j \in J_{a,r}} \end{matrix} ; \frac{z^a}{(a+1)^{a+1}} \right). \tag{4}$$

Indeed,

$$\prod_{j=1}^{a+1} \left(\frac{r+j}{a+1} \right)_n = \frac{((a+1)n+r)!}{r!(a+1)^{(a+1)n}},$$

and deleting the factor $(1)_n = n!$ yields (4).

To match the notation of the cited results, set

$$\rho_a := \frac{a}{(a+1)^{(a+1)/a}}, \qquad \lambda_j := \beta_j - 1,$$

and define

$$\varphi(w) := \sum_{n=0}^{\infty} \frac{(w/a)^{an}}{\prod_{j \in J_{a,r}} (\lambda_j + 1)_n}.$$

Then $F_{a,r}(z) = \varphi(\rho_a z)$. This is the $l = 0$, $t = a$ case of Salikhov's formula [7, eq. (1)]. Here $l = 0$ means that Salikhov's numerator contains no Pochhammer factors; in standard hypergeometric notation the parameter 1 in (4) remains because $(1)_n/n! = 1$.

Let $\theta = z\,d/dz$ and define

$$W_{a,r} := \operatorname{span}_{\mathbb{C}(z)}\{1, F_{a,r}, \theta F_{a,r}, \dots, \theta^{a-1} F_{a,r}\}. \tag{5}$$

The key functional statement is

$$\dim_{\mathbb{C}(z)} W_{a,r} = a + 1. \tag{6}$$

For odd a, Salikhov's Theorem 1 applies to this $l = 0$ function [7, Theorem 1]. The parameters are rational, and none is a negative integer. The remaining hypothesis requires that their multiset modulo integers not be invariant under translation by $1/d$ for any divisor $d > 1$ of a. In the present notation, this multiset is

$$S_a = \left\{ \frac{1}{a+1}, \dots, \frac{a}{a+1} \right\} \subset \mathbb{Q}/\mathbb{Z}.$$

If $d > 1$ divides a, then $\gcd(d, a+1) = 1$, so $1/d$ does not lie in the subgroup $(a+1)^{-1}\mathbb{Z}/\mathbb{Z}$ containing S_a. Translation by $1/d$ therefore cannot preserve S_a, which verifies Salikhov's parameter condition. The theorem gives algebraic independence of $\varphi(\alpha), \varphi'(\alpha), \dots, \varphi^{(a-1)}(\alpha)$ for every nonzero algebraic α.

When a is even, the theorem of Salikhov–Viskina applies with $t = a$. If the λ_j are rational and not negative integers, it states that for every nonzero algebraic α the numbers $1, \varphi(\alpha), \delta\varphi(\alpha), \ldots, \delta^{a-1}\varphi(\alpha)$ are linearly independent over $\overline{\mathbb{Q}}$, where $\delta = w\, d/dw$ is their Euler operator [8, p. 833]. Unlike Salikhov's odd-order result, this gives linear rather than algebraic independence; it includes the constant 1 and imposes no translation condition.

In either parity, the cited theorem implies that the values of $1, F_{a,r}, \theta F_{a,r}, \ldots, \theta^{a-1}F_{a,r}$ are linearly independent over $\overline{\mathbb{Q}}$ at every nonzero algebraic point. At such a point α, the Euler derivatives $(\theta^k F_{a,r})(\alpha)$ and the ordinary derivatives $F_{a,r}^{(k)}(\alpha)$ are related by a triangular algebraic matrix with diagonal entries $1, \alpha, \ldots, \alpha^{a-1}$. The two forms of the independence statement are therefore equivalent.

Descent lemma. *Let $g_1, \ldots, g_m \in \overline{\mathbb{Q}}[[z]]$. If these series are linearly dependent over $\mathbb{C}(z)$, then they are linearly dependent over $\overline{\mathbb{Q}}(z)$.*

Proof. Clear the denominators in a nonzero relation and let D be the largest degree of its polynomial coefficients. Writing those coefficients as $\sum_{d=0}^{D} c_{i,d} z^d$ and comparing Taylor coefficients gives a homogeneous linear system over $\overline{\mathbb{Q}}$ in the finitely many unknowns $c_{i,d}$. Its row space has a finite basis, so the system has the same kernel as a finite matrix over $\overline{\mathbb{Q}}$. If its kernel over $\mathbb{C}$ is nonzero, then its kernel over $\overline{\mathbb{Q}}$ is nonzero, and the corresponding polynomial coefficients give the required relation. $\square$

Every function in (5) has algebraic Taylor coefficients, so the descent lemma implies (6). Otherwise a relation over $\mathbb{C}(z)$ would descend to $\overline{\mathbb{Q}}(z)$. Specializing at a nonzero algebraic point where its coefficients are defined and not all zero would contradict the cited value independence.

Applying the value-independence statement at $z = 1$ proves that $F_{a,r}(1)$, and hence $C_{a,r}$ by (3), is transcendental for $0 \le r \le a$.

The same statement handles affine functions with negative intercept that remain positive for every $n \ge 1$. Suppose that $1 - a \le b < 0$ and put $r = a + 1 + b$, so that $a \ge 2$ and $2 \le r \le a$. Shifting the summation index by writing $n = m + 1$ gives

$$\begin{aligned} C_{a,b} &= \sum_{m=0}^{\infty} \frac{(m+1)!}{((a+1)m+r)!} \\ &= \frac{1}{r!}\left(F_{a,r}(1) + \frac{1}{a}\theta F_{a,r}(1)\right). \end{aligned}$$

If $C_{a,b}$ were algebraic, this identity would give a nontrivial linear relation over $\overline{\mathbb{Q}}$ among 1, $F_{a,r}(1)$, and $\theta F_{a,r}(1)$. Hence $C_{a,b}$ is transcendental for $1 - a \le b < 0$ as well.

To treat $b > a$, we compare the formal solutions of the relevant differential equations at infinity. The generalized hypergeometric equation for (4) can be written as

$$\left[\prod_{j \in J_{a,r}} \left(\frac{\theta}{a} + \beta_j - 1\right) - \frac{z^a}{(a+1)^{a+1}}\right] F_{a,r} = \prod_{j \in J_{a,r}} (\beta_j - 1). \tag{7}$$

The right side is nonzero because the parameter equal to 1 was removed. To describe the formal system at infinity, put

$$x := z^{-1}, \qquad K := \mathbb{C}((x)), \qquad \kappa_a := (a+1)^{-(a+1)},$$

so that $\theta = -x\, d/dx$ on K, and write

$$c_{a,r} := \prod_{j \in J_{a,r}} (\beta_j - 1), \qquad P_{a,r}(u) := \prod_{j \in J_{a,r}} (u + \beta_j - 1) = \sum_{j=0}^{a} p_j u^j, \qquad p_a = 1.$$

Let

$$Y_0 := 1, \qquad Y_j := \theta^{j-1} F_{a,r} \quad (1 \le j \le a).$$

Equation (7) is equivalent to the augmented companion system

$$\theta Y_0 = 0, \qquad \theta Y_j = Y_{j+1} \quad (1 \le j < a),$$

$$\theta Y_a = a^a c_{a,r} Y_0 + a^a(\kappa_a z^a - p_0) Y_1 - \sum_{j=1}^{a-1} a^{a-j} p_j Y_{j+1}.$$

After expanding rational functions at infinity, this system defines a rank-$(a+1)$ differential module $\mathcal{M}_{a,r}$ over K.

Lemma 4.1 (Zero-exponential formal component)**.** *The Levelt–Turrittin decomposition of $\mathcal{M}_{a,r}$ is*

$$\mathcal{M}_{a,r} = \mathcal{M}_0 \oplus \bigoplus_{\zeta^a = 1} \mathcal{M}_\zeta,$$

where $\mathcal{M}_0$ has rank one and exponential part 0, while each $\mathcal{M}_\zeta$ has rank one and exponential part $\rho_a \zeta z$. The zero summand has the formal solution vector

$$\widehat{Y}^{(0)} := \left(1, \widehat{F}^{(0)}_{a,r}, \theta \widehat{F}^{(0)}_{a,r}, \ldots, \theta^{a-1} \widehat{F}^{(0)}_{a,r}\right)^{\mathsf{T}},$$

where

$$\widehat{F}^{(0)}_{a,r}(z) \in z^{-a} \mathbb{C}[[z^{-a}]]. \tag{8}$$

More explicitly, if

$$G = r_0 + \sum_{k=0}^{a-1} r_{k+1} \theta^k F_{a,r} \in W_{a,r}, \qquad r_0, \ldots, r_a \in \mathbb{C}(z),$$

then its zero-exponential projection is

$$\pi_0(G) = r_0 + \sum_{k=0}^{a-1} r_{k+1} \theta^k \widehat{F}^{(0)}_{a,r} \in K.$$

The projection extends K-linearly to $K \otimes_{\mathbb{C}(z)} W_{a,r}$ and commutes with θ. In particular, $\pi_0(G)$ is a Laurent series without powers of $\log z$*.*

Proof. Writing $\widehat{F}^{(0)}_{a,r} = \sum_{m\geq 1} h_m z^{-am}$ in (7) gives

$$h_1 = -\frac{c_{a,r}}{\kappa_a}, \qquad h_{m+1} = \frac{P_{a,r}(-m)}{\kappa_a} h_m \quad (m \geq 1). \tag{9}$$

This recurrence constructs the solution (8). It is unique among Laurent-series particular solutions. Indeed, the difference of two such solutions would be a nonzero Laurent-series solution of the homogeneous equation. If dz^ν were its leading monomial, the term $-\kappa_a z^a dz^\nu$ could not be cancelled by $P_{a,r}(\theta/a)$, which does not raise exponents. Consequently $\widehat{Y}^{(0)}$ is a nonzero formal solution of the augmented system with exponential part 0.

The homogeneous equation associated with (7) has characteristic equation

$$\lambda^a = \frac{a^a}{(a+1)^{a+1}},$$

so its a characteristic numbers are the distinct nonzero numbers $\rho_a\zeta$. Its formal homogeneous solutions have the form

$$e^{\rho_a\zeta z} z^{\mu_\zeta}\mathbb{C}[[z^{-1}]], \qquad \zeta^a = 1,$$

and are log-free because every characteristic number is simple; see Salikhov [7, Remark 2, p. 206]. Each of the a distinct nonzero exponential parts therefore contributes a rank-one summand.

The Levelt–Turrittin theorem decomposes $\mathcal{M}_{a,r}$ as a direct sum of differential modules indexed by their exponential parts [9, Chapter 3, Theorem 3.1]. No ramification is needed because 0 and the polynomials $\rho_a\zeta z$ lie over K. The nonzero exponential summands account for rank a, and $\widehat{Y}^{(0)}$ supplies a nonzero solution in the zero-exponential summand. Since $\mathcal{M}_{a,r}$ has rank $a+1$, the zero summand has rank exactly one, and its formal solution space is spanned by $\widehat{Y}^{(0)}$.

By (6), the displayed expression for G in the statement is unique. Expanding the coefficients r_j in K and projecting the companion vector onto the line spanned by $\widehat{Y}^{(0)}$ gives the stated formula for $\pi_0(G)$. Because the Levelt–Turrittin summands are differential modules, this projection is K-linear and commutes with θ. Every entry of $\widehat{Y}^{(0)}$ lies in K, so the formula also shows directly that $\pi_0(G)$ contains no logarithms. $\square$

4.2 Euler reduction for $b > a$

Suppose that $b > a$ and write

$$b = q(a+1) + r, \qquad q \geq 1, \qquad 0 \leq r \leq a. \tag{10}$$

Put

$$\beta_j := \frac{b+j}{a+1}, \qquad 1 \leq j \leq a+1, \qquad x := \frac{z^a}{(a+1)^{a+1}}.$$

None of the lower parameters is equal to 1, and the multiplication formula gives

$$F_{a,b}(z) = {}_2F_{a+1}\left(\begin{matrix} 1, 1 \\ \beta_1, \ldots, \beta_{a+1} \end{matrix}; x\right). \tag{11}$$

The function $F_{a,b}$ in (11) is a hypergeometric E-function: its parameters are positive rational numbers, and with two upper and $a+1$ lower parameters, the required pullback degree is $(a+1)-2+1=a$. Writing its differential equation in terms of $\theta_x = x\,d/dx$, factoring off θ_x, and evaluating at $x=0$ to determine the resulting constant gives

$$\left[\prod_{j=1}^{a+1}\left(\frac{\theta}{a}+\beta_j-1\right)-\frac{z^a}{(a+1)^{a+1}}\left(\frac{\theta}{a}+1\right)\right]F_{a,b}=\prod_{j=1}^{a+1}(\beta_j-1). \tag{12}$$

This is an order-$(a+1)$ inhomogeneous equation. Its leading ordinary-derivative coefficient is a nonzero constant multiple of z^{a+1}, so its companion system has no nonzero finite singularity. Coefficient comparison in (2) gives the Euler reduction

$$\boxed{\prod_{k=1}^{q}\left(\frac{\theta}{a}+k\right)F_{a,b}(z)=\frac{b!}{r!}z^{-aq}\left[F_{a,r}(z)-r!\sum_{m=0}^{q-1}\frac{m!}{((a+1)m+r)!}z^{am}\right].} \tag{13}$$

The operator on the left multiplies the coefficient of z^{an} by

$$\prod_{k=1}^{q}(n+k)=\frac{(n+q)!}{n!}.$$

The index shift $n\mapsto n+q$ gives the right side of (13). In particular, the differential module generated by 1 and $F_{a,b}$ contains $W_{a,r}$.

4.3 Resonance

Assume for contradiction that $F_{a,b}\in W_{a,r}$ and apply the projection π_0 of Lemma 4.1 to (13). This projection commutes with the Euler operator on the left. On the right, the contribution of $z^{-aq}F_{a,r}$ begins at $z^{-a(q+1)}$ by (8), whereas the highest term of the polynomial subtracted in (13), corresponding to $m=q-1$, contributes the nonzero z^{-a} coefficient

$$-\frac{b!(q-1)!}{((a+1)(q-1)+r)!}. \tag{14}$$

Under this assumption, the lemma shows that the zero-exponential component of $F_{a,b}$ is a Laurent series $L(z)\in\mathbb{C}((z^{-1}))$ without logarithms. The Euler operator on the left of (13) acts diagonally on monomials, and its multiplier on z^{-a} is

$$\prod_{k=1}^{q}(-1+k)=0. \tag{15}$$

It follows that the z^{-a} coefficient in the zero-exponential component of the left side of (13) must vanish, contradicting (14). Therefore

$$F_{a,b}\notin W_{a,r}. \tag{16}$$

The projected equation therefore requires a logarithmic term. Since

$$\prod_{k=1}^{q}\left(\frac{\theta}{a}+k\right)\left(z^{-a}\log z\right)=\frac{(q-1)!}{a}z^{-a},$$

comparison with (14) gives

$$\widehat{F}^{(0)}_{a,b}(z)=-\frac{a\,b!}{((a+1)(q-1)+r)!}\,z^{-a}\log z+\cdots\,, \tag{17}$$

where the omitted terms do not affect the coefficient of $z^{-a}\log z$.

Define

$$V_{a,b}:=\operatorname{span}_{\mathbb{C}(z)}\{1,F_{a,b},\theta F_{a,b},\ldots,\theta^{a}F_{a,b}\}. \tag{18}$$

Equation (12) makes $V_{a,b}$ stable under θ and gives $\dim V_{a,b}\le a+2$. Equation (13) gives $W_{a,r}\subseteq V_{a,b}$. By (6) and (16), the additional element $F_{a,b}$ is not in $W_{a,r}$, so

$$\dim V_{a,b}\ge (a+1)+1=a+2.$$

Hence $\dim V_{a,b}=a+2$, and

$$1,F_{a,b},F'_{a,b},\ldots,F^{(a)}_{a,b} \tag{19}$$

are linearly independent over $\mathbb{C}(z)$. Thus (12) has minimal possible order among inhomogeneous equations over $\mathbb{C}(z)$ satisfied by $F_{a,b}$.

4.4 Values at $z=1$

By (11), $F_{a,b}$ is an E-function, as are its derivatives. The functions in (19) are linearly independent over $\overline{\mathbb{Q}}(z)$, and (12) supplies a companion system with no nonzero finite singularity. Beukers [6, Corollary 1.4] therefore gives the linear independence of their values at the ordinary algebraic point $z=1$. In particular, $F_{a,b}(1)$ is transcendental, and (3) gives the transcendence of $C_{a,b}$.

Combining this with the cases $1-a\le b\le a$ treated above proves Theorem 1. In terms of Problem 270, $S(f)$ is transcendental for every positive integer-valued affine function $f(n)=an+b$ with $a\ge 1$. Thus the theorem settles the entire affine subclass of the nondecreasing version of the problem.

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